Construction of the €STR Curve: OIS Bootstrap, ECB Meeting Dates and Proxy Quotes

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Abstract

This paper builds the €STR discounting curve, the risk-free curve of the euro area and the discount curve of collateralized euro derivatives. It extends the single-curve framework of the yield curve paper of this portfolio. We recall the definition of the €STR fixing and the cash flows of an overnight indexed swap (OIS), with daily compounding and market conventions (ACT/360, TARGET2 calendar, spot lag, payment lag), and we check them by rebuilding the ECB compounded €STR index from 1,797 daily fixings. The curve is bootstrapped from public data [1]; since €STR OIS quotes are not free, they are replaced by proxy quotes built from the ECB AAA government curve, shifted by the average gap between the AAA curve and the realized €STR (4.9 bp).

A standard bootstrap, with one pillar per OIS, reprices the 25 quotes to machine precision and matches QuantLib to 2×10^{-8} bp, but its overnight forwards jump at the OIS maturities or spread the policy decisions over several months. The short end is therefore rebuilt with pillars on the ECB effective dates, which gives piecewise constant overnight forwards and a readable policy path: on the proxy quotes, about 71 bp of hikes, i.e. 2.8 moves of 25 bp, are priced over 18 months. The two constructions agree on discounting within 2.3 bp of zero rate, but differ by 3 bp on the forward between two meetings. The implied path depends strongly on the proxy: one standard deviation of the AAA–OIS basis moves the number of hikes priced in one year from 2.4 to 3.1. Finally, the par-point risk of a forward-starting OIS shows that flat-forward curves give additive bucketed risks, whereas monotone convex does not. This paper is linked to my GitHub, which can be found on my portfolio[1], and all scripts are available in the same repository.

[1] https://portfolio-eliasbenguedda.fr

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1 Introduction

1.1 Motivation

To price vanillas, exotics, structured products, we need rates and so, first, we need to build a discount curve. Collateralized derivatives are discounted at the rate paid on the collateral, which for euro trades is the overnight rate €STR (Euro Short-Term Rate). Each IBOR tenor (EURIBOR 3M, 6M) has its own projection curve, and the basis spreads between them are material.

€STR is published by the ECB since 2 October 2019 and replaced EONIA (EONIA = €STR $+ 8.5$ bp until its discontinuation on 3 January 2022)[2]. Its OIS market is liquid from one week to 30 years. Building this curve is therefore the first step of any euro rates pricer:

(i) a *discount curve* for collateralized cash flows;

(ii) a *forward curve* of the overnight rate, whose short end reflects the expected ECB decisions;

(iii) the base on which the EURIBOR projection curves are built (Section 9).

The €STR curve is the risk-free zero-coupon curve of the euro area, i.e. a set of discount factors $P(0,t)$, from which zero rates and forwards are derived. Compared with the zero curve of the yield curve paper, it has three specificities: the underlying rate is an overnight rate compounded daily, its short end moves in steps with the ECB decisions, and the instruments follow precise market conventions (calendar, spot lag, payment lag). Since the €STR curve discounts the €STR flows themselves, the single-curve framework is exact here.

We must note that because OIS data is not available for free on internet we will use a proxy. We will take the AAA yield curve provided by the ECB as an approximation of the €STR curve, corrected by the average gap between the two curves (Section 4). The construction is independent of this choice: with market quotes, only the input file changes.

1.2 Organization of the paper

Section 2 defines the €STR fixing, the OIS cash flows and the conventions, and checks them against the ECB compounded index. Section 3 describes the data. Section 4 builds the proxy OIS quotes from the AAA curve. Section 5 performs a standard bootstrap, used as a baseline and to validate the instruments. Section 6 builds the short end on the ECB meeting dates and reads the implied policy path. Section 7 studies the par-point risk. Section 8 discusses the limitations, Section 9 lists next steps and Section 10 concludes. Appendix A gives excerpts of the Python implementation and Appendix B the notation.

[2]The equivalent of €STR in the US is SOFR, for the UK it is SONIA

2 The €STR Rate and Overnight Indexed Swaps

2.1 The €STR fixing

€STR is computed by the ECB from the overnight unsecured borrowing transactions of euro area banks with financial counterparties [6]. It is a volume-weighted trimmed mean: the transactions are ranked by rate, the top and bottom 25% of the volume are removed, and the remaining 50% are averaged with volume weights. It is published at 8:00 CET on the next TARGET2 business day for the previous day, together with the volume, the number of transactions and of reporting banks, and the share of the five largest banks; a contingency procedure applies when the data are insufficient.

The rate is an annualized simple rate on the ACT/360 basis. Investing one euro overnight from business day d_i to the next business day d_{i+1} pays $1 + r_i\, n_i/360$, where n_i is the number of calendar days (3 over a weekend).

2.2 Compounded €STR

Over an accrual period $[T_s, T_e]$ with business days $d_0 = T_s < d_1 < \cdots < d_N = T_e$, the floating coupon of an OIS is the daily compounded rate

$$R(T_s, T_e) = \frac{360}{D}\left(\prod_{i=0}^{N-1}\Big(1 + r_i\,\frac{n_i}{360}\Big) - 1\right), \qquad D = \sum_{i=0}^{N-1} n_i. \tag{1}$$

Under the risk-neutral measure associated with €STR discounting, $\prod_i (1 + r_i n_i/360)$ is the value at T_e of one euro rolled overnight from T_s. Its present value is therefore $P(0, T_s)$, so the floating leg of a period is worth

$$\mathrm{PV}_{\mathrm{float}} = P(0, T_s) - P(0, T_e), \tag{2}$$

without any convexity or timing adjustment, provided the payment date is T_e. With a payment lag, the period paid at $p_i > T_e$ is worth $P(0, p_i)\big(P(0, T_s)/P(0, T_e) - 1\big)$. Since the lag applies to both legs, its effect on the par rate is tiny: on our curve, removing the one-day lag changes the par rates by less than 0.0011 bp up to 30 years. It is nevertheless kept, so that the schedules match the market ones exactly (Section 5.4).

2.3 OIS par rate

A spot-starting OIS of maturity T exchanges annual fixed payments $K\,\delta_j$ (ACT/360) against the compounded €STR (1). With periods $[s_i, e_i]$ paid at p_i, the par rate is

$$K = \frac{\sum_i P(0, p_i)\big(P(0, s_i)/P(0, e_i) - 1\big)}{\sum_i \tau_i\, P(0, p_i)}, \qquad \tau_i = \frac{e_i - s_i}{360}, \tag{3}$$

which reduces to the formula of the yield curve paper without payment lag: $S(T) = \big(P(0, T_0) - P(0, T_n)\big)/\sum_j \delta_j P(0, t_j)$, T_0 being the spot date. OIS of maturity up to one year have a single payment at maturity: $S(T)\,\delta = P(0, T_0)/P(0, T) - 1$, the same equation as a deposit.

2.4 Conventions

2.5 Conventions check: rebuilding the ECB compounded index

The ECB publishes, with the fixings, a compounded €STR index $I_t = 100 \prod_k (1 + r_k n_k/360)$ (base 100 on 1 October 2019) and the compounded averages over 1 week to 12 months. Rebuilding them from the daily fixings checks the TARGET calendar (which business days have a fixing, and how many calendar days each fixing covers) and the ACT/360 day count. Over the 1,797 days of the sample, the rebuilt index differs from the ECB one by at most 5×10^{-9}, and

Item	€STR OIS convention
Calendar	TARGET2
Spot lag	$T + 2$ business days
Day count (both legs)	ACT/360
Fixed leg frequency	annual (short front stub; single payment up to 1 year)
Business day convention	modified following
Payment lag	1 business day
Fixing publication	$T + 1$, 8:00 CET
ECB decisions	effective on the Wednesday after the meeting
Model time	ACT/365 from the valuation date

Table 1: Conventions used in the implementation. The OIS schedules coincide date by date with those of QuantLib's `Estr` OIS (Section 5.4).

the averages agree to the fifth decimal (Table 2), which is the rounding of the published values (8 decimals for the index, 5 for the averages).

On 2026-10-06	Rebuilt from fixings	ECB	Difference
Index	109.935	109.935	-5.3×10^{-10}
1M average (%)	2.34817	2.34817	4.1×10^{-6}
3M average (%)	2.24781	2.24781	3.0×10^{-6}
6M average (%)	2.12073	2.12073	-4.4×10^{-6}
12M average (%)	2.04256	2.04256	-3.8×10^{-7}

Table 2: ECB compounded €STR index and averages rebuilt from the daily fixings.

These averages are realized, backward-looking rates. The curve is about the forward-looking expectation of the same compounded rates, which is what OIS quotes give; the index is used again in Section 4 to measure the gap between the AAA curve and the €STR.

3 Data

3.1 €STR fixings and the ECB corridor

The data are downloaded from the ECB Data Portal [7]: 1,797 daily €STR fixings from 1 October 2019 to 6 October 2026, the ECB key rates, the meeting calendar and the AAA government curve. The valuation date is 5 October 2026, with an €STR of 2.438% and a deposit facility rate of 2.50%.

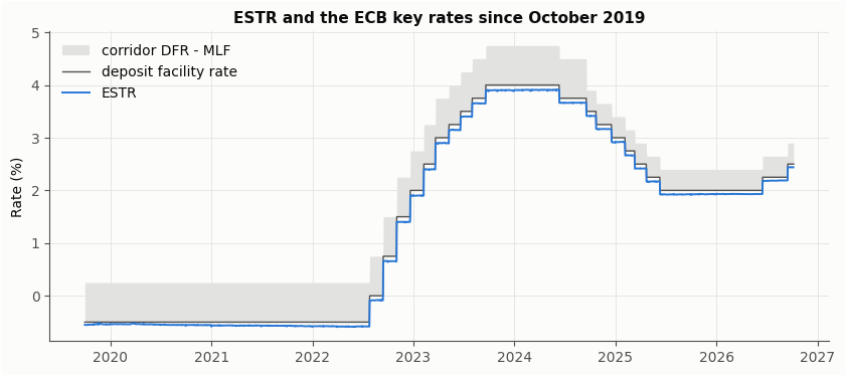


Figure 1: €STR fixings and ECB key rates from October 2019 to October 2026.

Figure 1 also shows the ECB key rates. The deposit facility rate (DFR) is the rate the ECB pays to banks on their overnight deposits at the central bank. In the current regime of excess liquidity it is the main policy rate: a bank has no reason to lend overnight below the DFR, since it can deposit its cash at the ECB without any risk. The marginal lending facility rate (MLF), at which banks can borrow overnight from the ECB against collateral, is the ceiling of the corridor, and the main refinancing rate (MRO) lies between the two. The corridor DFR–MLF[3] therefore bounds the overnight rates of the banks.

The €STR is nevertheless *below* the DFR, because it includes the transactions with non-bank lenders (money market funds, insurers, public institutions) that have no access to the deposit facility and accept to lend to banks below the DFR.

3.2 The €STR – DFR spread

Table 3 gives the spread between the €STR and the DFR by year. It is negative, between −4 bp and −10 bp, and very stable within a year: its standard deviation is below 1 bp, while the dispersion of the transactions around the fixing (25–75 percentile range) is of 3 to 6 bp. The spread widened until 2023 with the excess liquidity and narrows since then as the liquidity is withdrawn (−6.7 bp on average in 2026, −6.3 bp over the last three months).

This stability is what makes the policy path the backbone of the short end of the curve: between two meetings the €STR is almost constant, and it jumps on the effective date of each decision (Section 6).

3.3 AAA government curve (source of the proxy quotes)

The data are in the `ECB_AAA_YieldCurve.csv` file, which also contains the Svensson parameters published by the ECB. Table 4 gives the spot rates of the valuation date.

[3] DFR–MLF: Deposit Facility Rate – Marginal Lending Facility Rate

Year	Mean (bp)	Std (bp)	Min (bp)	Max (bp)	25–75 range (bp)
2019	−4.2	0.7	−5.5	−1.1	3.9
2020	−4.7	0.9	−8.3	−2.2	3.4
2021	−6.8	0.5	−9.0	−5.8	4.3
2022	−8.6	0.8	−11.0	−7.2	3.8
2023	−10.0	0.4	−12.0	−9.3	6.0
2024	−8.9	0.5	−11.3	−8.1	4.9
2025	−7.8	0.5	−9.4	−6.6	4.9
2026	−6.7	0.3	−7.4	−5.8	4.3

Table 3: €STR − DFR spread by year, and mean 25–75 percentile range of the transactions.

Tenor	Rate (%)	Tenor	Rate (%)	Tenor	Rate (%)
3M	2.5305	4Y	3.1072	10Y	3.5020
6M	2.6743	5Y	3.1674	12Y	3.6079
9M	2.7780	6Y	3.2349	15Y	3.7237
1Y	2.8527	7Y	3.3051	20Y	3.8178
2Y	2.9971	8Y	3.3745	25Y	3.8229
3Y	3.0555	9Y	3.4406	30Y	3.7722

Table 4: ECB AAA government spot rates (continuously compounded) on 2026-10-05.

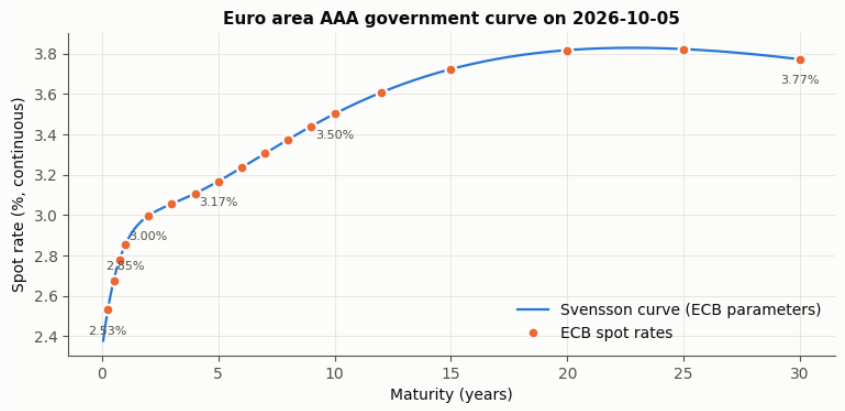


Figure 2: Euro area AAA government curve on 2026-10-05: Svensson curve and ECB spot rates.

3.4 ECB monetary policy meetings

We also imported the list of the Governing Council monetary policy meeting dates. This data is important because, as can be seen on the €STR fixings, the rate moves in a discretionary manner: rates are decided during the ECB meetings, and the new rates are effective on the Wednesday of the following week, at the start of the reserve maintenance period. After the valuation date, the calendar contains 18 effective dates up to 13 December 2028, 16 of them within two years; the first ones are 4 November 2026, 23 December 2026, 10 February 2027 and 24 March 2027.

3.5 From the €STR fixing to a curve

One may wonder how a whole curve can be built from a single overnight rate. The €STR fixing only gives one point: the overnight rate of today. The information on future rates comes from overnight indexed swaps. An OIS of maturity T exchanges a fixed rate against the €STR compounded up to T, so its par rate reflects the path of the €STR that the market expects until T, plus a term premium. Since the €STR closely tracks the ECB deposit facility rate, this expected path is mainly the expected path of monetary policy. Bootstrapping the OIS par rates then gives the discount factors of the €STR curve, and hence its zero rates and forwards. In this paper, market OIS quotes are replaced by the proxy quotes of Section 4, so the curve reflects the expectations embedded in the AAA government curve rather than those of the OIS market.

4 Proxy OIS Quotes

4.1 Svensson curve

The ECB fits the Svensson model [10] on the prices of the AAA-rated euro area government bonds, and publishes in `ECB_AAA_YieldCurve.csv` both the fitted parameters and the spot rates of the fitted curve. The continuously compounded spot rate of maturity T is

$$z_{\mathrm{AAA}}(T) = \beta_0 + \beta_1 \left(\frac{1 - e^{-T/\tau_1}}{T/\tau_1} \right) + \beta_2 \left(\frac{1 - e^{-T/\tau_1}}{T/\tau_1} - e^{-T/\tau_1} \right) + \beta_3 \left(\frac{1 - e^{-T/\tau_2}}{T/\tau_2} - e^{-T/\tau_2} \right). \quad (4)$$

The formula defines the curve at every maturity, not only at the published tenors, which is what the proxy needs: its instantaneous forward $f_{\mathrm{AAA}}(0,T) = \partial_T (T\, z_{\mathrm{AAA}}(T))$ is available in closed form, and par rates can be computed on any dated schedule.

Tenor	ECB spot (%)	Svensson (%)	Diff. (bp)	Tenor	ECB spot (%)	Svensson (%)	Diff. (bp)
3M	2.530476	2.530476	-3.8×10^{-10}	6Y	3.234878	3.234878	7.8×10^{-9}
6M	2.674286	2.674286	3.2×10^{-9}	7Y	3.305120	3.305120	6.2×10^{-9}
9M	2.778026	2.778026	3.4×10^{-9}	8Y	3.374513	3.374513	7.7×10^{-9}
1Y	2.852743	2.852743	4.9×10^{-9}	9Y	3.440616	3.440616	9.0×10^{-9}
2Y	2.997130	2.997130	5.3×10^{-9}	10Y	3.501981	3.501981	3.9×10^{-9}
3Y	3.055535	3.055535	2.7×10^{-9}	12Y	3.607910	3.607910	8.1×10^{-9}
4Y	3.107177	3.107177	2.2×10^{-9}	15Y	3.723674	3.723674	3.1×10^{-9}
5Y	3.167448	3.167448	-5.0×10^{-10}	20Y	3.817818	3.817818	5.0×10^{-9}
				25Y	3.822865	3.822865	2.3×10^{-10}
				30Y	3.772198	3.772198	7.7×10^{-10}

Table 5: ECB AAA spot rates (continuously compounded) on 2026-10-05 against the Svensson formula evaluated with the ECB parameters ($\beta_0 = 1.7202$, $\beta_1 = 0.6117$, $\beta_2 = 1.9283$, $\beta_3 = 6.7187$, $\tau_1 = 0.9659$, $\tau_2 = 13.6006$). Difference = ECB − Svensson, in basis points.

Since the published spot rates are themselves computed by the ECB from the fitted curve, Table 5 is not a test of the model but of our implementation: the differences, below 10^{-8} bp, are the rounding of the published values, and they confirm the parameter conventions (rates in percent, τ_1, τ_2 in years). Two limits of the AAA curve should be kept in mind: it is itself the output of a smooth fit, which does not pass exactly through the bond prices, and no AAA bond is shorter than about one year, so its short end is an extrapolation.

4.2 Government vs OIS basis

The AAA curve is not the €STR curve: the two rates do not carry the same risk and are not used for the same purpose. The ECB AAA curve is fitted on the central government bonds of the AAA-rated euro area countries, mainly Germany and the Netherlands. These bonds are the euro area safest assets: they are used as collateral and as a store of value, so their yield contains a convenience and liquidity premium, and it also depends on the supply of public debt and on repo specialness. An OIS, on the other hand, only exchanges a fixed rate against the compounded overnight rate and carries almost no credit or liquidity premium. The difference between the two curves, the government–OIS basis, therefore depends on the maturity and on the market regime, and it can change sign. During stress periods, the flight to quality pushes AAA yields below OIS rates.

Since €STR OIS quotes are not available, the basis cannot be read directly from the market. We measure it ex post, by replacing the OIS rate with what the €STR actually did: on each date t, the 3-month AAA spot rate is compared with the €STR compounded over the following three months, computed from the ECB index I_t,

$$b(t) = z_{\mathrm{AAA}}(t; 3\mathrm{M}) - \frac{1}{\tau} \ln \frac{I_{t+3\mathrm{M}}}{I_t}, \quad (5)$$

where both rates are continuously compounded and τ is the ACT/365 length of the period. Figure 3 shows the result.



Figure 3: AAA 3M rate minus €STR compounded over the next 3 months (ex post).

The basis is a few basis points below zero in calm periods, but it falls to about -100 bp in 2022. This measure must be read with care, for three reasons:

- it mixes the basis with forecast errors: the realized €STR includes the policy surprises, and in 2022 the ECB hiked much faster than expected, so most of the 2022 trough is an error of anticipation, not a difference of risk;
- no AAA bond is as short as three months, so the 3-month point of the Svensson curve is an extrapolation: the basis is measured on the least reliable part of the AAA curve;
- it is only measured at three months, whereas the basis depends on the maturity.

Over the last two years, the basis has a mean of $\bar{b} = -4.9$ bp and a standard deviation of 8.9 bp. The mean is used to build the proxy quotes, and the standard deviation gives the order of magnitude of their uncertainty (Section 6.7).

4.3 Proxy forward curve and quotes

The proxy instantaneous forward curve is built in two parts:

$$f^{\text{proxy}}(0,t) = \begin{cases} r_{ON}, & t < E_1, \\ f_{\text{AAA}}(0,t) - \bar{b}, & t \geq E_1, \end{cases} \tag{6}$$

where $r_{ON} = 365\ln(1 + r_0/360)$ is today's €STR fixing r_0 converted to a continuously compounded ACT/365 rate, E_1 is the first ECB effective date (4 November 2026) and $f_{\text{AAA}}(0,t)$ is the instantaneous forward of the Svensson curve (4). Before E_1 no decision can change the €STR, so the forward is set to today's fixing. Beyond, the proxy is the AAA forward shifted up by $-\bar{b} = 4.9$ bp, the average gap between the realized €STR and the AAA curve. The forward jumps at E_1, which is consistent with a policy decision on that date. The integrated forward $y(t) = \int_0^t f^{\text{proxy}}$ is computed on a fine grid (600 points up to 32 years), so the interpolation of the proxy curve plays no role.

The proxy quotes are the par rates (3) of 25 dated OIS, from 1 week to 30 years, priced on this curve with the conventions of Table 1 (Table 6). By construction, the quotes below three months are model-generated: the 1W and 2W OIS end before E_1 and are simply equal to today's fixing, while the 1M and 2M OIS combine the fixing with the extrapolated AAA forward. When a quote file `ESTR_OIS_Quotes_YYYYMMDD.csv` is filled, these market quotes replace the proxy and the rest of the construction is unchanged.

Tenor	Par rate (%)	Tenor	Par rate (%)	Tenor	Par rate (%)
1W	2.4385	1Y	2.9090	10Y	3.5281
1M	2.4502	2Y	3.0521	15Y	3.7221
3M	2.5654	3Y	3.1091	20Y	3.8046
6M	2.7118	5Y	3.2178	30Y	3.7966

Table 6: Proxy €STR OIS par rates on 2026-10-05 (selection of the 25 tenors; annual ACT/360 fixed leg against compounded €STR, spot start 2026-10-07).

5 Standard Bootstrap: Baseline and Validation

5.1 Role of the standard bootstrap

Before building the short end on the ECB meeting dates, the curve is first bootstrapped in the standard way, with one pillar per OIS quote, following the approach of the yield curve paper [9]. This curve is not the final €STR curve, but it serves three purposes.

(i) **Validation.** This is the first time the dated OIS instruments (TARGET calendar, $T+2$ spot lag, ACT/360 accruals, front stub, payment lag) are run through the bootstrap. Since the standard bootstrap reprices every quote exactly, any residual pricing error points to a problem in the instrument definitions or the solver. It therefore validates the whole chain, from instruments to bootstrap to curve. The meeting-step curve of Section 6 cannot serve this purpose, since it does not reprice the short-end quotes exactly.

(ii) **Long end.** Beyond the last ECB meeting date, the meeting-step curve of Section 6 is extended with one pillar per remaining OIS quote and flat forwards between pillars, i.e. with the same method as the raw bootstrap of this section.

(iii) **Benchmark.** Building both curves from the same quotes shows why the short end requires the ECB meeting dates, and quantifies the impact of this choice (Section 6).

The interpolation methods themselves (locality, smoothness, linearity) are compared in the yield curve paper [9]. Only two of them are retained here. The first is the *raw* method, which interpolates log discount factors linearly and therefore produces flat forwards between pillars; since the interpolated curve depends linearly and locally on the pillar values, its risk sensitivities stay local and stable. The second is the *monotone convex* method of Hagan and West [2], which yields continuous, positive forwards and is widely used in practice.

5.2 Instruments and pillars

Each OIS of Table 6 is an instrument whose implied quote is the par rate (3), computed from the curve on its dated schedule. One pillar is placed at the last payment date of each OIS, measured in ACT/365 from the valuation date, and the unknowns are the continuously compounded zero rates at the 25 pillars. An OIS only depends on the curve up to its last payment, so each new instrument adds exactly one unknown and the system is square.

The intermediate dates of a long OIS do not all fall on pillars: the period end dates e_i are one business day before the payment dates, and beyond 10 years the annual payment dates (11, 13, 14 years…) have no quote of their own. The discount factors at these dates are interpolated and depend on the pillar being solved, so the par-rate equation (3) has no closed-form solution for the last discount factor and is solved numerically. With raw interpolation the system is triangular: the curve up to a pillar only depends on the previous pillars, and each pillar is solved in turn with a one-dimensional root search (Brent). With monotone convex a new pillar also moves the curve before the previous ones, and the 25 equations are solved simultaneously (Powell hybrid method, starting from the raw solution).

5.3 Results

Both methods reprice the 25 quotes to machine precision (Table 8). The simultaneous solve of monotone convex is about a hundred times slower than the sequential raw bootstrap: every evaluation of the residuals rebuilds the curve and reprices the dated OIS, and the Jacobian is obtained by finite differences. This is acceptable for one curve a day, but not for a risk engine that rebuilds the curve once per bumped quote.

Instrument	Quote	Pricing equation
€STR fixing (overnight)	rate r_0	$1 + r_0\, n_0/360 = 1/P(0, d_1)$
Short OIS ($\leq$ 1Y)	rate S	$1 + S\,\delta = P(0, T_0)/P(0, T)$
OIS ($>$ 1Y)	par rate K	equation (3)
Meeting-date OIS $[E_k, E_{k+1}]$	forward rate F	$1 + F\,\delta_k = P(0, E_k)/P(0, E_{k+1})$

Table 7: Instruments of the €STR curve (payment lag omitted in the equations). Only standard OIS are used with the proxy quotes; meeting-date OIS would be added with market quotes.

Method	Max residual (bp)	Solver	Time (ms)
Raw	1.1×10^{-12}	sequential (Brent)	≈ 40
Monotone convex	6.6×10^{-9}	simultaneous (hybr)	$\approx 4\,300$

Table 8: Standard bootstrap on the 25 proxy OIS quotes.

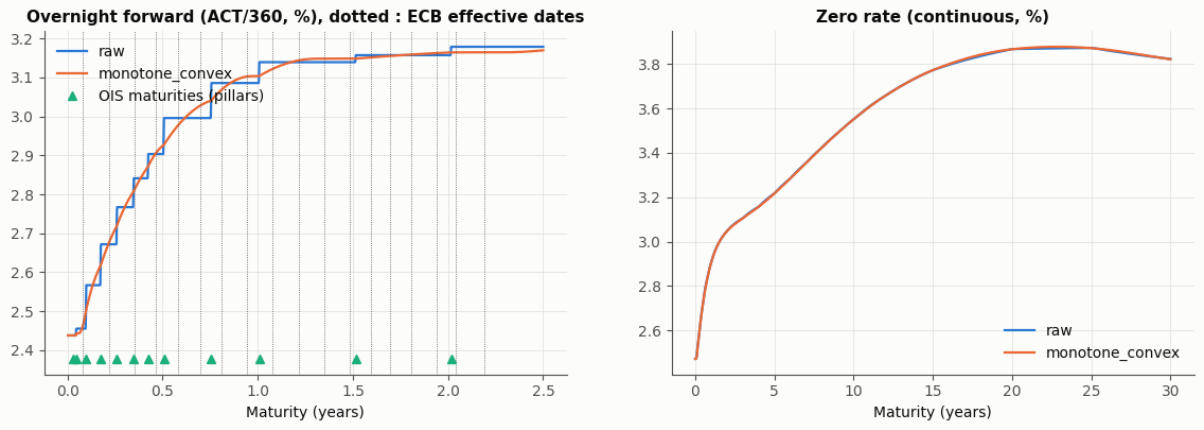


Figure 4: Standard bootstrap. Left: overnight forward up to 2.5 years, with the ECB effective dates (dotted) and the OIS maturities (triangles). Right: zero rates.

On the zero rates the two curves are indistinguishable (Figure 4, right). The overnight forwards, however, show the problem of the short end (left):

- **raw** jumps at the OIS maturities, i.e. at the pillars (triangles), dates on which nothing happens to the €STR;
- **monotone convex** moves continuously and spreads each ECB decision over several months.

Neither reflects how the €STR actually moves: in steps, on the ECB effective dates (dotted lines). Both curves even let the forward rise before the first effective date, when no decision can change the €STR. This is the motivation of Section 6, which keeps the raw bootstrap for the long end and replaces the short end by steps on these dates.

5.4 Validation of the bootstrap

Repricing the input instruments with the functions used to build the curve is necessary but not sufficient, since an error shared by the bootstrap and the pricer (schedule, accrual, lag) cancels out. The curve is validated by the following tests.

(1) **Repricing.** All quotes are repriced with a maximum residual of 1.1×10^{-12} bp (raw) and 6.6×10^{-9} bp (monotone convex), below the tolerance of 10^{-6} bp.

(2) **Independent implementation (QuantLib [8]).** Two checks, with the same conventions (`Estr` index, `MakeOIS` / `OISRateHelper`, $T + 2$, payment lag of one business day):

- our curves are passed to QuantLib as discount curves, and QuantLib prices the 25 OIS: its fair rates differ from our implied quotes by at most 2×10^{-13} bp. The schedules coincide date by date (for instance the 18M OIS pays on 8 April 2027 and 10 April 2028 in both implementations), which validates the calendar, the stub, the day count and the payment lag;
- QuantLib bootstraps its own curve from the same quotes with log-linear discount factors, the equivalent of raw (`PiecewiseLogLinearDiscount`): over 30 years, its zero rates differ from ours by at most 2×10^{-8} bp.

(3) **Exact round trip.** Quotes are generated from a known curve whose pillars are the instrument maturities, then bootstrapped back with the same interpolation: the curve is recovered to 5×10^{-11} bp at the pillars and 1.4×10^{-10} bp on the instantaneous forwards over 30 years, for both methods.

(4) **Sanity checks.** The discount factors decrease strictly from 0.99993 (one day) to 0.318 (30 years), and the instantaneous forwards stay between 2.47% and 4.25% (continuously compounded, ACT/365; 2.47% is today's €STR of 2.438% ACT/360 in this convention).

(5) **Robustness.**

- The result does not depend on the order of the instruments (the bootstrapper sorts them by maturity).
- Starting the monotone convex solve from the raw solution shifted by ± 50 bp or $+200$ bp leads to the same curve within 2×10^{-7} bp. The solver does not report convergence at its very strict tolerance (10^{-15}), which is why the bootstrap is checked on the residuals, not on the solver flag.
- A parallel $+1$ bp shift of all quotes moves the zero curve by between 0.98 and 1.01 bp: par rates and zero rates do not move one for one.

- The sum of the par-point sensitivities equals the parallel sensitivity for raw, but not for monotone convex, which is not linear in its inputs (Section 7).

(6) **Out of sample (leave-one-out).** Each inner quote is removed in turn, the curve is rebuilt on the 24 others and the removed quote is repriced. The error has a root mean square of 1.42 bp for raw (maximum 3.6 bp on the 1M) and 0.32 bp for monotone convex (maximum 1.0 bp on the 1M). This favours monotone convex by construction, since the proxy quotes are generated from a smooth curve; with market quotes, which price a step path on the short end, the ranking could be different. The test comparing an OIS quote with the €STR realized afterwards requires historical OIS quotes and is left for real data.

Test	Criterion	Result
Repricing	$\max \lvert R_k \rvert < 10^{-6}$ bp	1.1×10^{-12} (raw), 6.6×10^{-9} (MC) bp
QuantLib pricing of our OIS	identical par rates	$\leq 2 \times 10^{-13}$ bp
QuantLib bootstrap	identical curve	$\leq 2 \times 10^{-8}$ bp on zero rates
Exact round trip	machine precision	$\leq 1.4 \times 10^{-10}$ bp
Discount factors, forwards	decreasing, plausible	0.318–0.99993; 2.47–4.25%
Order / starting point	identical curve	identical; $\leq 2 \times 10^{-7}$ bp
Parallel +1 bp	≈ 1 bp	0.98–1.01 bp
Leave-one-out	within bid-ask	rms 1.42 bp (raw), 0.32 bp (MC)

Table 9: Validation tests of the standard €STR bootstrap (MC: monotone convex).

6 The Short End on ECB Meeting Dates

6.1 Construction

The overnight rate only moves materially on the ECB decision effective dates, and stays close to the deposit facility rate in between (Table 3). A natural model of the short end is therefore a piecewise constant overnight forward with jumps on the effective dates $E_1 < E_2 < \ldots$:

$$f(0,t) = \begin{cases} r_{ON}, & t < E_1, \\ f_k, & t \in [E_k, E_{k+1}), \end{cases} \tag{7}$$

with $f(0,\cdot)$ piecewise constant and $f(0,t)$ the instantaneous forward rate at time t, the pillars being placed on the meeting dates rather than on the instrument maturities. On $[0, E_1)$ the forward is today's fixing, known since no decision can change it before E_1. The jumps $f_k - f_{k-1}$ are the market-implied policy moves, and $(f_k - f_{k-1})/25$ bp the implied number of 25 bp cuts or hikes.

The nodes are the effective dates up to a horizon of two years: 16 dates, from 4 November 2026 to 13 September 2028, hence 15 unknown levels $f_1, \ldots, f_{15}$. The integrated forward at the nodes is

$$y(E_1) = r_{ON} E_1, \qquad y(E_{k+1}) = y(E_k) + f_k\,(E_{k+1} - E_k), \tag{8}$$

and the curve between the nodes is the raw interpolation of the yield curve paper: linear in $y = -\ln P$, hence exactly the flat forwards of (7). No new interpolation is needed: only the position of the pillars changes.

6.2 More meetings than quotes: penalty and choice of λ

Only 11 OIS (1W to 18M) mature before the last node, for 15 unknown levels: the system is underdetermined, and infinitely many step paths reprice the short quotes. The levels are therefore obtained by least squares on the residuals (in basis points), with a penalty on the jumps,

$$\min_{f_1,\ldots,f_{15}} \sum_{k \in \text{short}} R_k^2 + \lambda^2 \sum_{k=1}^{15} (f_k - f_{k-1})^2, \qquad f_0 = r_{ON}, \tag{9}$$

with $\lambda = 0.002$ (residuals and jumps in basis points). Among the paths that reprice the quotes, the penalty selects the one with the smallest jumps. Table 10 shows the effect of λ:

- with $\lambda = 0$ the problem is ill-posed: the optimizer returns an oscillating path with 3,675 bp of cumulated jumps and -31.5 moves priced in one year;
- for λ between 10^{-4} and 10^{-2} the solution does not change: the penalty only selects a path, and the residual of 0.79 bp does not come from λ (Section 6.4);
- from $\lambda \approx 0.1$ the fit is traded for smoothness, and at $\lambda = 1$ the curve no longer reprices the market (5.4 bp of error).

The value $\lambda = 0.002$ lies in the neutral region. A leave-one-out on the short quotes gives an optimum around $\lambda = 0.03$ (root mean square error 0.52 bp against 0.57 bp at $\lambda = 0.002$), a small difference which is moreover biased towards smoothing, since the proxy quotes come from a smooth curve. With market quotes, λ would rather be chosen by the discrepancy principle: the largest value for which every quote stays repriced within its bid–ask.

The penalty also has a side effect: the levels beyond the last short maturity (18 months, April 2028) are constrained by no quote, so the penalty sets them equal to the previous one. The last four levels of Table 12 (March to July 2028) are therefore identical, a plateau that is an artefact of the regularization, not market information.

λ	Max residual (bp)	RMS residual (bp)	$\sum \lvert f_k - f_{k-1} \rvert$ (bp)	25bp moves in 1 year
0	0.785	0.254	3,675	−31.5
10^{-4} to 10^{-2}	0.785	0.254	72	2.72
0.03	0.797	0.257	72	2.70
0.1	0.870	0.289	72	2.70
0.3	1.089	0.550	71	2.68
1	5.358	2.564	65	2.42

Table 10: Effect of the penalty λ on the fit of the 11 short OIS and on the implied path.

6.3 Long end

Beyond the last node, each remaining OIS (2Y to 30Y) adds one pillar at its last payment date, with a flat forward from the previous pillar, solved sequentially with a one-dimensional root search: this is exactly the raw bootstrap of Section 5. The whole curve is a single `YieldCurve` with raw interpolation on the union of the pillars (effective dates, then OIS maturities), and can be used directly by the pricers of the other papers.

6.4 Residuals

The long quotes are repriced exactly (below 10^{-11} bp). The short ones are not quite: 0.79 bp on the 1M, −0.27 bp on the 3M, 0.12 bp on the 4M, and less than 0.04 bp elsewhere. The 1M (end on 9 November) and the 2M (end on 7 December) OIS both end between the first two effective dates, and the forward before E_1 is fixed at today's €STR: a single level f_1 has to fit two quotes. The system is thus underdetermined overall but overdetermined locally. Proxy quotes generated from a smooth curve are not exactly compatible with a step path; market quotes, which already price a step path, would give much smaller residuals, and the remaining ones would measure the turn-of-month and turn-of-year effects that a richer curve would add as extra steps.

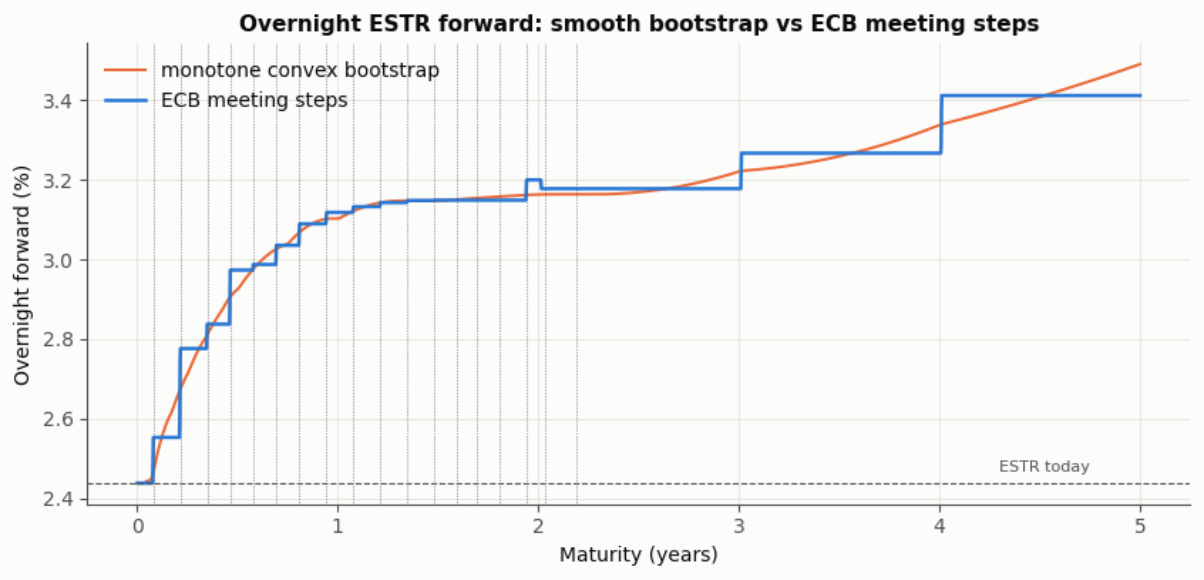


Figure 5: Overnight €STR forward: monotone convex bootstrap vs ECB meeting steps.

6.5 What changes with respect to the standard bootstrap

Both constructions use the same quotes and differ only by where the overnight forward changes level: at the OIS maturities for the standard bootstrap, at the ECB effective dates for the meeting steps. Table 11 reads the overnight forward on a few dates around the first two effective dates.

Overnight forward (%) on	Raw	Monotone convex	Meeting steps	
20/10/2026	2.438	2.438	2.438	today's €STR
03/11/2026	2.456	2.460	2.438	day before E_1
06/11/2026	2.456	2.476	2.553	after E_1 (4/11)
12/11/2026	2.567	2.516	2.553	after the 1M pillar
10/12/2026	2.672	2.627	2.553	after the 2M pillar
28/12/2026	2.672	2.690	2.776	after E_2 (23/12)
Max $\lvert\Delta r\rvert$ vs steps, 0–2 years (bp)	2.32	1.87	–	
Max $\lvert\Delta P\rvert$ vs steps, 0–2 years	5.0×10^{-5}	4.0×10^{-5}	–	
Forward $E_1 \to E_2$ (%)	2.590	2.585	2.557	

Table 11: Standard bootstrap vs meeting steps on the same quotes (overnight forwards in ACT/360).

Up to the first effective date the €STR cannot change: the meeting steps keep today's fixing, while raw and monotone convex already drift up. After it, the steps jump once and stay flat until the next decision, whereas raw jumps at the 1M and 2M payment dates. Discounting is barely affected (at most 2.3 bp on zero rates and 5×10^{-5} on discount factors up to two years), but the forward between the first two meetings differs by about 3 bp with the same quotes: the construction matters for forward-starting instruments and for reading the policy path, not for plain discounting.

6.6 Implied ECB policy path

Each level f_k is the €STR expected after a decision. Converted to an overnight ACT/360 rate and diminished by the current €STR − DFR spread (-6.3 bp over the last three months), it gives the implied deposit facility rate, and its change since today in units of 25 bp moves (Table 12).

Decision	Effective	€STR forward (%)	Implied DFR (%)	Change (bp)	25bp moves
29/10/2026	04/11/2026	2.553	2.616	11.5	0.46
17/12/2026	23/12/2026	2.776	2.839	33.8	1.35
04/02/2027	10/02/2027	2.838	2.900	40.0	1.60
18/03/2027	24/03/2027	2.974	3.036	53.6	2.14
29/04/2027	05/05/2027	2.988	3.051	55.0	2.20
10/06/2027	16/06/2027	3.036	3.099	59.8	2.39
22/07/2027	28/07/2027	3.090	3.153	65.2	2.61
09/09/2027	15/09/2027	3.118	3.181	68.0	2.72
28/10/2027	03/11/2027	3.133	3.196	69.5	2.78
16/12/2027	22/12/2027	3.143	3.206	70.5	2.82
03/02/2028	09/02/2028	3.148	3.211	71.0	2.84
23/03/2028	29/03/2028	3.149	3.212	71.1	2.85
04/05/2028	10/05/2028	3.149	3.212	71.1	2.85
08/06/2028	14/06/2028	3.149	3.212	71.1	2.85
20/07/2028	26/07/2028	3.149	3.212	71.1	2.85

Table 12: Implied €STR forwards and deposit facility rate between ECB meeting dates (proxy quotes). Change and moves are measured from today's €STR of 2.438%.

On the proxy quotes, about 34 bp of hikes are priced by the end of 2026 and 71 bp, i.e. 2.8 moves of 25 bp, over 18 months; the moves are fractional because the forwards are expectations over the possible decisions. Three caveats apply. First, with proxy quotes this is the path implied by the AAA government curve, not by the OIS market. Second, risk-neutral forwards include a term premium, so they are not a forecast of the decisions. Third, the plateau of 2028 is set by the penalty, not by the data.

6.7 Sensitivity to the proxy assumption

The proxy depends on the AAA–OIS basis $\bar{b}$. Table 13 rebuilds the quotes and the meeting-step curve with the basis shifted by one standard deviation, and without basis.

Basis $\bar{b}$	2Y OIS (%)	5Y OIS (%)	10Y OIS (%)	25bp moves in 1 year
−13.8 bp (mean −1 std)	3.1388	3.3065	3.6170	3.07
−4.9 bp (2-year mean)	3.0521	3.2178	3.5281	2.72
+4.0 bp (mean +1 std)	2.9654	3.1291	3.4392	2.37
0 (no basis)	3.0040	3.1686	3.4788	2.53

Table 13: Proxy quotes and implied path for different AAA–OIS basis assumptions.

One standard deviation of the basis moves every OIS rate by about 9 bp and the number of moves priced in one year from 2.4 to 3.1. The basis acts almost as a parallel shift of the forwards beyond E_1, but since the first level is anchored on today's fixing, a parallel shift of the forwards becomes a change of the size of the first decisions. The construction is independent of the proxy; its levels are not, and market quotes are needed before reading anything economic into them.

7 Risk: Par-Point Sensitivities

The instrument is an 18-month × 4-year forward-starting receiver OIS: it receives a fixed rate of 3% on a notional of EUR 10 million, from 18 months to 5.5 years, so its cash flows lie between the pillars. Its value on the meeting-step curve is −127,475 EUR, since the forward OIS rate is above 3%. Each quote is bumped by +1 bp in turn, the curve rebuilt with the same construction and the trade repriced (Table 14, Figure 6).

Quote bumped	Meeting steps	Monotone convex	Raw
6M	−8	−7	−7
1Y	30	31	31
18M	1,429	1,436	1,427
2Y	33	15	34
3Y	6	−12	6
4Y	14	225	14
5Y	−2,270	−2,794	−2,270
6Y	−2,762	−3,940	−2,762
7Y	0	403	0
8Y	0	2	0
Sum of par points	−3,528	−4,641	−3,528
Parallel +1 bp	−3,526	−3,527	−3,526

Table 14: Par-point DV01 (EUR) of an 18m × 4y receiver OIS, EUR 10m (quotes with |DV01| > 1 EUR).

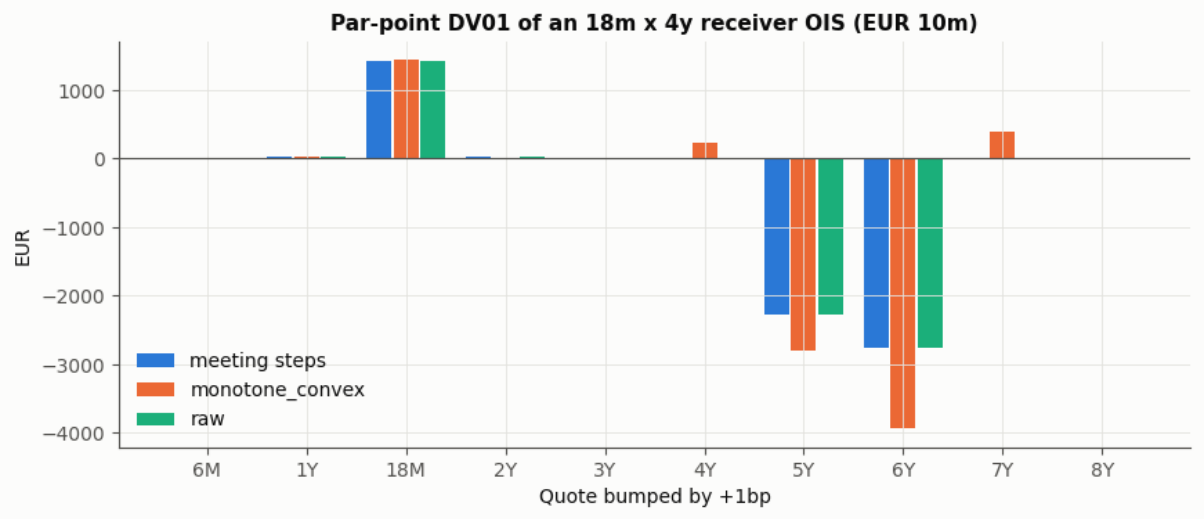


Figure 6: Par-point DV01 of an 18m × 4y receiver OIS (EUR 10m).

The parallel DV01 is the same for the three constructions, about −3,527 EUR per basis point: a receiver loses when rates rise. Its split across the quotes shows three things.

- The risk sits on the quotes that bracket the flows: the 18M quote, at the start of the swap (+1,429 EUR: a higher 18M rate lowers $P(0, 18\text{M})$ and the value of the floating leg that the receiver pays), and the 5Y and 6Y quotes, around its end at 5.5 years (−2,270 and −2,762 EUR).
- **Meeting steps and raw give the same risk.** Both have flat forwards between their pillars, and the flows of the trade start after the last meeting node (18 months), where both curves are built the same way.

- **Monotone convex spreads the risk** on the 4Y and 7Y quotes (+225 and +403 EUR), and its par-point DV01s add up to −4,641 EUR against a parallel DV01 of −3,527 EUR. The method is not linear in its inputs (its forward constraints switch regions when a quote moves, as in the yield curve paper), so its bucketed risk is not additive. This is a practical argument for flat forwards on a risk curve, on top of the monetary policy argument of Section 6.

Bumping and rebuilding costs one bootstrap per quote; the bootstrap Jacobian, $\partial z/\partial q = -(\partial_z R)^{-1}\partial_q R$, would give all the sensitivities at once (Section 9).

8 Limitations

Proxy quotes. The levels of the curve come from the AAA government curve, shifted by a basis measured ex post at three months only, which mixes the basis with forecast errors and has varied by tens of basis points since 2019 (down to -100 bp in 2022). The same constant basis is applied at every maturity, and the quotes below three months are generated by the model. One standard deviation of the basis moves the OIS rates by 9 bp and the policy path by 0.35 moves in one year.

Data. The curve is built on a single valuation date, from mid quotes; the bid–ask spread, which would be the natural tolerance of the fit and of the choice of λ, is ignored.

Conventions. The conventions match QuantLib's €STR OIS, but the holiday calendar is stored from 2019 to 2060 only, and the rounding rules of the compounded rate in real confirmations are not reproduced.

Short end. The meeting-step model assumes that the €STR does not move between two decisions, and that the €STR $-$ DFR spread is constant; turn-of-month and turn-of-year effects are not modelled. The choice of λ and of the two-year horizon is ad hoc, and the levels beyond the last short quote are set by the penalty.

Long end. The forward is flat beyond the last quote (30 years). Long-dated €STR OIS are also less liquid than EURIBOR swaps, so with real data the long end would rather be built from EURIBOR swaps and €STR/EURIBOR basis swaps.

Collateral. €STR discounting assumes cash collateral in euros remunerated at €STR; other collateral agreements (USD collateral, no CSA) require other discount curves.

9 Next Steps

- Use real quotes: fill `ESTR_OIS_Quotes_YYYYMMDD.csv`, and add meeting-date OIS, which pin down each level of the short end directly.
- Choose λ and the horizon of the meeting nodes by the discrepancy principle, with the residuals normalized by the bid–ask of each quote.
- Gather the code in an `EstrCurveBuilder` class (TARGET calendar, €STR index, instruments, quote source, builder) returning a `YieldCurve`, with the policy path and the par-point risk as methods.
- Compute the risk with the bootstrap Jacobian instead of bumping and rebuilding.
- Build the EURIBOR 3M and 6M projection curves on €STR discounting (multi-curve framework).
- Calibrate Hull–White on €STR discounting and price the callable range accrual and Bermudan swaptions.
- Rebuild the curve every day and study its principal components, as in the yield curve paper.
- Cross-currency: SOFR curve and EUR/USD basis.

10 Conclusion

The €STR curve is the discount curve of collateralized euro derivatives. This paper builds it from public data, with the market conventions of €STR OIS. The conventions are validated twice: the ECB compounded index is rebuilt from 1,797 fixings to 5×10^{-9}, and the OIS priced on our curve match QuantLib to 2×10^{-13} bp, with identical schedules.

A standard bootstrap, with one pillar per OIS, reprices the 25 quotes exactly and is independently reproduced by QuantLib, but its overnight forwards do not move like the €STR: they jump at the OIS maturities (raw) or spread each decision over several months (monotone convex). Placing the short-end pillars on the ECB effective dates gives a step path, flat between decisions, from which the expected policy can be read directly: on the proxy quotes, 2.8 hikes of 25 bp are priced over 18 months. Both constructions discount almost identically, within 2.3 bp of zero rate, but differ by 3 bp on a forward between two meetings, and the flat-forward constructions give additive bucketed risks, which monotone convex does not.

The main limit is the data: without free OIS quotes, the levels of the curve come from the AAA government curve and an estimated basis, whose uncertainty of one standard deviation moves the implied path by 0.35 hikes in one year. The construction itself is independent of this proxy, and with market quotes only the input file changes.

A Python Implementation (Excerpts)

The following listings are excerpts of the implementation (package `Fixed_Income_Models` and notebook `estr_curve.ipynb`), slightly simplified. The full scripts and the notebook are available on GitHub.

A.1 Dated €STR OIS

```
class OIS(Instrument):
    # Spot-starting ESTR OIS with a dated schedule ; maturity (pillar) = last
        payment time
    def __init__(self, tenor, rate, spot=SPOT, pay_lag=1):
        end = add_tenor(spot, tenor)
        months = ...                                    # tenor in months (0 for
            weeks)
        ends, k = [end], 12
        while months - k > 0:                           # annual periods, short
            front stub
            ends.insert(0, modified_following(spot + pd.DateOffset(months=
                months - k)))
            k += 12
        starts = [spot] + ends[:-1]
        pays = [roll(e, pay_lag) for e in ends]         # payment 1 business day
            after each end
        self.t_start, self.t_end, self.t_pay = time(starts), time(ends), time(
            pays)
        self.accrual = np.array([(e - s).days / 360 for s, e in zip(starts,
            ends)])
        super().__init__(self.t_pay[-1], rate)

    def implied_quote(self, curve):
        P_pay = curve.discount(self.t_pay)
        floating = np.sum(P_pay * (curve.discount(self.t_start) / curve.
            discount(self.t_end) - 1))
        return floating / np.sum(self.accrual * P_pay)
```

A.2 Proxy quotes

```
def proxy_curve(basis_bp=BASIS_BP):
    # ESTR flat until the first effective date, then AAA forward minus the
        basis
    grid = np.unique(np.concatenate([[FIRST_EFFECTIVE], np.linspace(
        FIRST_EFFECTIVE, 32, 600)]))
    h = 1e-4
    aaa_y = lambda t: svensson(t, *row[SVENSSON]) / 100 * t
    fwd = (aaa_y(grid + h) - aaa_y(grid - h)) / (2 * h) - basis_bp * 1e-4
    y = ON * FIRST_EFFECTIVE + np.concatenate([[0.0], np.cumsum(0.5 * (fwd[1:]
        + fwd[:-1]) * np.diff(grid))])
    return YieldCurve(grid, y / grid, "linear_zero")

quotes = [OIS(t, 0.0) for t in TENORS]
for q in quotes:
    q.quote = q.implied_quote(proxy_curve())
```

A.3 Meeting-step curve

```
def meeting_curve(quotes, effective_times, horizon=2.0, penalty=0.002):
    nodes = np.array([e for e in effective_times if e <= horizon])        # E_1
        < ... < E_n
```

```
short = [q for q in quotes if q.maturity <= nodes[-1]]
long = [q for q in quotes if q.maturity > nodes[-1]]

def build(levels):
    # forward ON on [0, E_1), levels[k] on [E_{k+1}, E_{k+2}) : raw
        interpolation of y
    y = ON * nodes[0] + np.concatenate([[0.0], np.cumsum(levels * np.diff(
        nodes))])
    return YieldCurve(nodes, y / nodes, "raw"), y

def residuals(levels):
    c, _ = build(levels)
    fit = [1e4 * q.residual(c) for q in short]
    jumps = penalty * 1e4 * np.diff(np.concatenate([[ON], levels]))
    return np.concatenate([fit, jumps])

sol = least_squares(residuals, np.full(len(nodes) - 1, ON))
_, y = build(sol.x)
times, zeros = list(nodes), list(y / nodes)
for q in long:                                    # long end : one pillar
    per OIS, flat forward
    f = lambda z: q.residual(YieldCurve(np.array(times + [q.maturity]), np.
        array(zeros + [z]), "raw"))
    zeros.append(brentq(f, -0.05, 0.2, xtol=1e-14)); times.append(q.
        maturity)
return YieldCurve(np.array(times), np.array(zeros), "raw"), pd.Series(sol.x
    , index=nodes[:-1]), short
```

B Notation

Symbol	Meaning
$r_i,\ n_i$	€STR fixing of business day d_i and number of calendar days to the next business day
$R(T_s, T_e)$	Daily compounded €STR over $[T_s, T_e]$
I_t	ECB compounded €STR index
$P(0,T),\ y(T)$	€STR discount factor and integrated forward $-\ln P(0,T)$
T_0	Spot date (valuation date +2 business days)
$s_i,\ e_i,\ p_i,\ \tau_i$	Start, end and payment dates of period i, ACT/360 accrual
$S(T),\ K$	€STR OIS par rate
$r_0,\ r_{ON}$	Today's €STR fixing, and its continuously compounded ACT/365 equivalent
$z_{\text{AAA}},\ f_{\text{AAA}}$	AAA Svensson spot rate and instantaneous forward
$b(t),\ \bar{b}$	Ex-post AAA–€STR basis and its 2-year mean
$E_k,\ f_k$	ECB decision effective dates and overnight forwards on $[E_k, E_{k+1})$
λ	Penalty on the jumps of the meeting-step forwards
$q_k,\ R_k,\ z$	Market quotes, residuals and bootstrap unknowns

Table 15: Main notation.

References

[1] ECB data, *https://data.ecb.europa.eu*

[2] P. S. Hagan and G. West, *Interpolation methods for curve construction*, Applied Mathematical Finance, 13(2):89–129, 2006.

[3] F. M. Ametrano and M. Bianchetti, *Everything you always wanted to know about multiple interest rate curve bootstrapping but were afraid to ask*, SSRN Working Paper 2219548, 2013.

[4] M. Henrard, *Interest Rate Modelling in the Multi-Curve Framework*, Palgrave Macmillan, 2014.

[5] L. B. G. Andersen and V. V. Piterbarg, *Interest Rate Modeling, Volume I: Foundations and Vanilla Models*, Atlantic Financial Press, 2010.

[6] European Central Bank, *Euro short-term rate (€STR) methodology and policies*, Available at: https://www.ecb.europa.eu/stats/financial_markets_and_interest_rates/euro_short-term_rate/html/index.en.html

[7] European Central Bank, *ECB Data Portal*, Available at: https://data.ecb.europa.eu

[8] *QuantLib, a free/open-source library for quantitative finance*, Available at: https://www.quantlib.org

[9] E. Benguedda, *Yield Curve Construction: Interpolation, Bootstrapping and Curve Risk*, 2026. Available on the portfolio: https://portfolio-eliasbenguedda.fr

[10] L. E. O. Svensson, *Estimating and interpreting forward interest rates: Sweden 1992–1994*, NBER Working Paper 4871, 1994.